

Multiple Choice (10 marks)

1. Which of the following is usually NOT represented in a subroutine's activation record frame for a stack-based programming language?
 - (a) Values of local variables
 - (b) A heap area
 - (c) The return address
 - (d) Stack pointer for the calling activation record
2. Which of the following does authentication aim to accomplish?
 - (a) Restrict what operations/data the user can access
 - (b) Determine if the user is an attacker
 - (c) Flag the user if he/she misbehaves
 - (d) Determine who the user is
3. Which of the following language features requires that stack-based storage allocation be used rather than static allocation?
 - (a) Recursive procedures
 - (b) Arbitrary goto's
 - (c) Two-dimensional arrays
 - (d) Integer-valued functions
4. Which of the following is not a sentence that is generated by the grammar $A \rightarrow BC$, $B \rightarrow x|Bx$, $C \rightarrow B|D$, $D \rightarrow y|Ey$, $E \rightarrow z$?
 - (a) xyz
 - (b) xy
 - (c) xxzy
 - (d) xxxxy
5. How does a buffer overflow on the stack facilitate running attacker-injected code?
 - (a) By overwriting the return address to point to the location of that code
 - (b) By writing directly to the instruction pointer register the address of the code
 - (c) By writing directly to %eax the address of the code
 - (d) By changing the name of the running executable, stored on the stack

True / False (10 marks)

Answer True or False

6. True or False: WPA primarily relies on AES encryption to secure wireless communications.
7. True or False: To verify a digital signature, one must use the sender's public key.
8. True or False: A proxy firewall filters traffic primarily at the application layer of the OSI model.
9. True or False: WPA 3 provides the strongest wireless security among the available Wi-Fi security protocols.
10. True or False: Availability is not considered one of the classic security properties in information security.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to drop empty items from a given dictionary.
12. Write a function to split a string at uppercase letters.

End of Paper